Charles Parker

Biostatistician

charlesnp98@gmail.com | *(757)-373-3783* | *linkedin.com/in/charlesnparker*

Summary

Biostatistician with an M.S. in Biostatistics and experience applying statistical models for genomic and clinical data. Research interests include statistical methodology, machine learning, and applications to large-scale high-dimensional data.

Education

Oregon Health and Science University *Sep 2024 - Jun 2026*

Master of Science (MS) | Biostatistics | GPA: 3.94

Portland State University *Jan 2023 - Jun 2024*

Bachelor of Science (BS) | Inter-Disciplinary Science | GPA: 3.92

Minors: Mathematics

Tidewater Community College *Jan 2020 - Dec 2020*

Associate's Degree | Science | GPA: 4.0

Work Experience

Surgical Technologist *Nov 2017 – Jan 2023*

Naval Medical Center Portsmouth | United States Navy, Portsmouth, VA | 40 Hours/week

- Promoted to Oral Maxillofacial Clinic's Assistant Manager in 2021, overseeing 15 personnel, coordinating surgical supplies, maintaining clinical safety procedures in accordance with Naval Medical and HIPPA standards, and providing continued high-quality care to patients
- Standardized on-boarding procedures that improved workflow consistency and supported a 100% successful on-boarding record over four years
- Managed competing clinical priorities while maintaining accuracy, meeting deadlines, and supporting efficient daily clinic operations in a high-volume surgical environment

Research Experience

Graduate Researcher *Dec 2024 - Jun 2026*

Karstens Lab | OHSU, Portland, OR | 20 Hours/week

- Planned, coordinated, and executed multiple quantitative research studies using clinical, genomic, and patient survey data to support evidence-based research decision
- Applied generalized linear models, mixed-effects models, logistic regression, probability-based methods, and predictive modeling to evaluate complex biomedical research questions and support scientific decision-making
- Collected, integrated, cleaned, validated, and analyzed high-dimensional clinical and biological datasets using reproducible analytical methods

Biomedical Informatics and Data Science Intern *May 2024 - Aug 2024*

Karstens Lab | OHSU, Portland, OR | 40 Hours/week

- Developed and optimized computational analysis scripting that improved sequencing workflow and reduced computation time by 15 hours
- Presented project objectives, methodology, analytical findings, and recommendations to faculty researchers and laboratory leadership

Teaching Experience

Graduate Teaching Assistant *Sep 2025 - Apr 2026*

OHSU-PSU School of Public Health | Portland, OR | 20 Hours/week

- Supported graduate-level instruction in probability and statistical inference through weekly office hours and individualized mentoring
- Provided guidance on interpreting results and statistical analysis for large multi-part projects

Biostatistics Tutor *Sep 2025 - Jun 2026*

OHSU-PSU School of Public Health | Portland, OR | 20 Hours/week

- Assisted cross-disciplinary students from OHSU in probability and statistical inference
- Communicated complex statistical concepts through clear explanations and practical examples; improving student understanding and research competency

Selected Research Projects

Implementing General Linear Mixed Effect Model Using Urinary Microbiome Data *Apr 2026*

Charles Parker

- Adapted a Linear Mixed-Effects modeling framework for urinary microbiome data, using Zero-Inflated Poisson and Negative Binomial Models to account for low taxa read counts
- Compared model results with traditional microbiome analyses and identified significant taxa between microbiome groups

Incorporate Covariates into GLM for UroMicrobiome Analysis *Apr 2025*

Charles Parker

- Compared traditional, quantitative microbiome profiling (QMP), and QMP with gene copy number (GCN) abundance methods while incorporating covariates to control variation between pre- and post-menopausal groups.
- Evaluated how abundance measurement methods affected downstream statistical analysis and detection of differences in the urinary microbiome.

Prediction Model for Overactive Bladder Using Clinical Characteristics Data *Sep 2025*

Charles Parker

- Developed logistic regression models to predict wet versus dry overactive bladder using clinical and urinary microbiome characteristics, incorporating diversity and bacterial abundance metrics.
- Applied LASSO for variable selection and explored LOCOM for differential abundance analysis to identify factors associated with overactive bladder classification.

Publications

Does Quantitative Microbiome Profiling Alter Interpretation of the Urinary Microbiome *May 2025*

Charles Parker, Christina Turner, Sean Davin, Benji Lamp, Keisha Harrison, PhD, Lisa Karstens, PhD

- Presented a poster from Karstens Lab as First Author at NW Women's Health & Sex/Gender Differences Research Conference
- Compared traditional and quantitative microbiome profiling (QMP) methods using diversity and differential-abundance analyses of urinary microbiome data from pre- and post-menopausal women.
- Evaluated alpha/beta diversity and differential abundance using statistical testing and QMP produced similar diversity results but altered the identification and interpretation of significant taxa, including Lactobacilli.

Skills

Statistical Programming: R, Python, SAS, STATA, SQL

Machine Learning: LASSO, Random Forest, GLM Predictive Modeling

Quantitative Analysis: Advanced Statistical Modeling, Regression Analysis, Statistical Inference

Data Management: Data Cleaning, Data Validation, Exploratory Data Analysis, Reproducible Workflows

Technical Competencies: Linux, Technical Reports, Git, Git Hub, AWS, Google Cloud Platform, Jupyter Notebook

Awards

Summa Cum Laude *Jun 2024*

Portland State University

CMC Challenge Coin *Aug 2019*

Command Master Chief, USNS Comfort

Armed Forces Service Medal *Nov 2019*

United State Navy